

Electrical Circuits Glossary and Basic Rules

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Glossary of Terms

Basic Rules for Solving Circuits

Ohm's Law

Ohm's Law states that the voltage (V) across a resistor is proportional to the current (I) through the resistor, given by $V = IR$. This is fundamental to solving circuits.

Kirchhoff's Voltage Law (KVL)

Kirchhoff's Voltage Law states that the sum of all the voltages around any closed loop in a circuit must equal zero. For example, in a simple series circuit:

$$V_{\text{battery}} = V_1 + V_2 + V_3$$

where V_1 , V_2 , and V_3 are the voltage drops across the resistors.

Kirchhoff's Current Law (KCL)

Kirchhoff's Current Law states that the total current entering a junction must equal the total current leaving the junction:

$$I_{\text{in}} = I_{\text{out}}$$

This is useful for analyzing parallel circuits.

Power in a Circuit

Power in an electrical circuit is given by $P = VI$ and can also be expressed as:

$$P = I^2 R \quad \text{or} \quad P = \frac{V^2}{R}$$

Power represents the rate at which energy is consumed by a resistor or other component.

Solving Circuits with Batteries, Switches, and Resistors

1. **Step 1**: Identify all resistors, power sources (batteries), and switches.
2. **Step 2**: Use Ohm's Law $V = IR$ to calculate unknown values (current, resistance, voltage).
3. **Step 3**: Apply Kirchhoff's Voltage Law (KVL) to any closed loops to find unknown voltages.
4. **Step 4**: Apply Kirchhoff's Current Law (KCL) to any junctions to find unknown currents.
5. **Step 5**: Calculate the power consumed by each resistor using $P = I^2 R$.